

HOME PLOTS APPS LIVE EDITOR INSERT VIEW Search (Ctrl+Shift+Space)

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FILE NAVIGATE TEXT CODE ANALYZE SECTION RUN

C:\Users\ADMIN\Documents\MATLAB

Files

- 41.mlx
- Addition and multiplication....
- causal non causal 42.mlx
- continuous time system.mlx
- Discrete.mlx
- Energy abd power 32.mlx
- evenodd.mlx
- Periodic.mlx

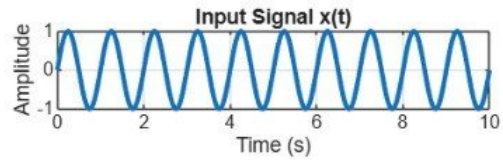
Workspace

Name	Value	Size	Class
t	1×1001 dou...	1×1001	double
x	1×1001 dou...	1×1001	double
x_delay	1×1001 dou...	1×1001	double
y	1×1001 dou...	1×1001	double

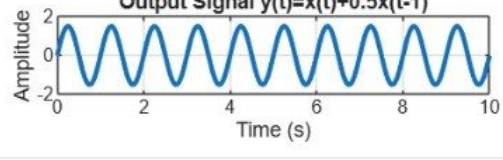
continuous time system.mlx

```
1 % Analysis of Continuous-Time System
2 clc;
3 clear;
4 close all;
5 % Time axis
6 t = 0:0.01:10;
7 % Input signal
8 x = sin(2*pi*t);
9 % Delayed signal x(t-1)
10 x_delay = sin(2*pi*(t-1));
11 % System output
12 y = x + 0.5*x_delay;
13 % Plotting
14 figure;
15 subplot(2,1,1);
16 plot(t, x, 'b', 'LineWidth', 2);
```

Input Signal x(t)



Output Signal y(t)=x(t)+0.5x(t-1)



Command Window

New to MATLAB? See resources for [Getting Started](#).

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FILE NAVIGATE

Text Normal B I U M

CODE

Code Issues Debugger

Section Break Run and Advance Run to End

Run Step Stop

ANALYZE SECTION RUN

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Workspace

Name	Value	Size	Class
a	1	1×1	double
b	[1,0.5000]	1×2	double
n	1×11 double	1×11	double
x	1×11 double	1×11	double
y	1×11 double	1×11	double

C:\Users\ADMIN\Documents\MATLAB\step response of system.mlx

```

11 a = [1];
12 % Step response
13 y = filter(b,a,x);
14 % Display result
15 disp('Step Response y(n):');
16 disp(y);
17 % Plot
18 figure;
19 subplot(2,1,1);
20 stem(n,x,'filled');
21 grid on;
22 title('Unit Step Input u(n)');
23 xlabel('n');
24 ylabel('Amplitude');
25 subplot(2,1,2);
26 stem(n,y,'filled');

```

Command Window

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Step Response y(n):

1.0000 1.5000 1.5000 1.5000 1.5000

